Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/30/2015
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Development of Multiple Antibiotic Resistance by Opportunistic Bacterial Pathogens during Human Space Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	WLN@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-261-3773
Organization Name:	University of Florida		
PI Address 1:	Space Life Sciences Laboratory		
PI Address 2:	505 Odyssey Way, Room 201-B		
PI Web Page:			
City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/27/2012	**End Date:**	09/30/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 9/30/2015 per NSSC information (Ed., 4/14/15) NOTE: New end date of 4/30/2015 (original end date was 7/26/13), per NSSC information (Ed., 4/14/14) NOTE: new end date of 4/30/2014 (original end date was 7/26/13), per NSSC information (Ed., 8/20/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fajardo-Cavazos, Patricia (University of Florida)		
Grant/Contract No.:	NNX12AN70G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Background: Maintaining astronaut health during long-term space flight is of prime importance to the success of exploration missions to destinations such as the Moon, near-Earth asteroids, or Mars. The immune system weakens during missions in Low Earth Orbit (LEO), rendering astronauts more susceptible to infection. Microgravity induces in certain bacterial species increased virulence properties such as: adherence to mammalian cells, biofilm formation, resistance to acid and macrophages, and antibiotic resistance. As part of the normal population of microbial inhabitants on and within the human body, astronauts carry many microbes considered “opportunistic pathogens”, i.e., normally harmless microbes with the potential to cause disease in hosts with lowered immune function. Opportunistic infections are often treated with combinations of two antibiotics that differ in their mechanisms of action. Combination treatment can be effective because (i) certain antibiotic combinations work synergistically to more effectively combat the pathogen than either drug given alone, and (ii) the probability of the pathogen evolving resistance to two antibiotics simultaneously is much lower than that of becoming resistant to a single antibiotic.
Rationale: Susceptibility of potential pathogens to single antibiotics in the space flight environment has been studied to a limited extent, but a systematic study of the development of multiple antibiotic resistance by mutation has not to date been attempted. In a previous space exposure experiment (PROTECT on International Space Station-ISS using the bacterium Bacillus subtilis), we found that certain mutations to rifampicin (RIF) resistance also resulted in resistance to trimethoprim (TMP). RIF and TMP are prescribed in combination for treatment of certain infections caused by the important opportunistic pathogen Staphylococcus aureus.

Goal: We propose to test the hypothesis that exposure to the human space flight environment, particularly microgravity, increases the development of simultaneous resistance to RIF and TMP. As our test organisms we will use the reference B. subtilis strain 168 and Staphylococcus epidermidis strain ATCC 12228, a harmless human commensal organism closely related to S. aureus.

Central Objectives:

1. Measure mutation rates of B. subtilis and S. epidermidis to RIF and TMP resistance, singly and in combination, in space flight and ground control samples.

2. Recover antibiotic-resistant mutants and determine the nucleotide sequence changes in the rpoB and dfrA genes leading to resistance to RIF and TMP, respectively.

3. Characterize altered phenotypes of the mutants obtained in Objectives (1) and (2).

Methods: Preliminary ground-based studies will be performed using rotating High Aspect Ratio Vessels (HARVs) to simulate microgravity. Space flight experiments will be performed using the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFUs) developed for use on the International Space Station with asynchronous ground controls. HARVs, BRIC-PDFUs, and support personnel are conveniently located at the Space Life Sciences Lab, Kennedy Space Center. All microbiology and molecular biology techniques proposed are routinely performed in our laboratory.

Significance: The proposed study responds to the National Research Council's (NRC’s) 2011 Decadal Report Recommendation P2, to: “Establish a robust space flight program of research analyzing plant and microbial growth and physiological responses to the multiple stimuli encountered in space flight environments,” and directly relates to section C2 of the present NASA Research Announcement (NRA)-“Rapid Turnaround Space Flight Experiments,” specifically addressing the question “What are the genetic changes associated with responses to space flight in biological systems?” Understanding multidrug resistance in opportunistic pathogens during long-term human space flight directly benefits astronaut health and the success of extended missions both in LEO and in interplanetary space.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

In several ways, astronauts on long-duration space flights share commonalities with patients in long-term clinical care settings: Both groups are relatively immobile, immunocompromised, share close living quarters with others in a similar weakened physiologic condition, and are susceptible to opportunistic infections. Understanding how multiple antibiotic resistance arises in opportunistic human pathogens during space flight can be directly translated to Earth-based medicine.

Final Progress Report, NASA grant NNX12AN70G:
“Development of multiple antibiotic resistance by opportunistic bacterial pathogens during human space flight”

Year 1:

1. Preparation for BRIC-18 mission to ISS. In Year 1 we took considerable effort to prepare for the Dragon III mission to ISS. With the help of NASA and the BRIC payload development team we successfully passed SVT and proceeded to PVT (now known as Experiment Verification Testing, or EVT)

2. Ground simulation studies. Using Synthecon Rotary Cell Culture Systems, we performed experiments to test the hypothesis that exposure of bacterial cells to simulated microgravity increases the frequency of spontaneous mutations to antibiotic resistance. We used two clinostats each equipped with four 10-ml High Aspect Ratio Vessels (HARVs), one placed in the Vertical (V) position to simulate microgravity and one placed in the Horizontal (H) position as a 1xg control. We showed that B. subtilis and S. epidermidis cells grew at equal exponential rates in the V and H positions, but in cell viability decline in stationary phase was significantly slower in the V orientation.

After 6 days of cultivation, B. subtilis cells exhibited a significantly higher frequency of mutation to rifampicin (RFM) resistance in the V orientation. This observation supports our Central Hypothesis. A variable percentage of RFM mutants also become resistant to trimethoprim (TMP).We initiated the same studies above using S. epidermidis, but suffered a slight setback when we discovered that the gas exchange membranes in four of our HARVs had developed pinhole leaks. These HARVs were returned to Synthecon for repair and returned in good condition

3. Acquisition and training of NASA-SAIP undergraduate intern Raed Narvel. Through a competitive selection procedure, we attracted Raed Narvel to participate in this project. Raed, a senior undergraduate Microbiology & Cell Science major at University of Florida (UF), was awarded a 10-week summer internship from the NASA-Florida Space Grant Consortium Space Aeronautics Internship Program. Raed proved to be a very bright intern and his efforts were instrumental in successfully completing the ground-based clinostat studies.

Task Progress:	Year 2: 1. Conclusion of pre-flight Ground Simulation studies. Using Synthecon Rotary Cell Culture Systems, we performed experiments to test the hypothesis that exposure of bacterial cells to simulated microgravity increases the frequency of spontaneous mutations to antibiotic resistance. We used two clinostats each equipped with four 10-ml High Aspect Ratio Vessels (HARVs), one placed in the Vertical (V) position to simulate microgravity and one placed in the Horizontal (H) position as a 1xg control. We showed that both Bacillus subtilis and Staphylococcus epidermidis cells grew at equal exponential rates in the V and H positions, but that cell viability decline in stationary phase was significantly slower in the V orientation in both organisms. After 6 days of cultivation, B. subtilis cells exhibited a significantly higher frequency of mutation to rifampicin (RFM) resistance in the V orientation, supporting the Central Hypothesis. A variable percentage of RFM mutants also become resistant to trimethoprim (TMP). However, S. epidermidis cells demonstrated an equivalent frequency of mutation to RFMR in either the H or V orientation, failing to support the Central Hypothesis. Thus, the two model organisms behaved differently under ground-based microgravity simulations. 2. Preparation and launch of BRIC-18 mission to ISS. Our experiments passed Science Verification testing (SVT) and Payload Verification Testing (PVT) and all NASA Flight Readiness Reviews. Various issues led to repeated delays of the launch of BRIC-18 to ISS. Finally, BRIC canisters were integrated and successfully launched on the SpaceX 3rd cargo delivery mission to the ISS on April 18, 2014. Year 3: Objective 1. Publication of pre-flight Ground Simulation studies. We published our pre-flight ground simulation studies in the journal Gravitational and Space Research 2(2): 34-45. 2014. Below is reprinted the Abstract: Bacteria of the genera Bacillus and Staphylococcus are frequent inhabitants of the International Space Station (ISS) and represent possible opportunistic pathogens. The effect of simulated microgravity on growth and the frequency of mutation to antibiotic resistance in the model surrogate organisms Bacillus subtilis (B. subtilis) and Staphylococcus epidermidis (S. epidermidis) were investigated. The test organisms were cultivated for six days in Rotating Wall Vessel (RWV) clinostats either in the vertical (simulated microgravity) or horizontal (1 g control) orientation. Parameters measured were: optical densities (ODs); viable counts; frequencies of resistance to rifampicin (RFM); and frequencies of double resistance to RFM and trimethoprim (TMP). The results indicated that the response to simulated microgravity differed in the two microorganisms. Both B. subtilis and S. epidermidis grew to higher ODs and cell numbers in simulated microgravity. However, the frequencies of mutation, both to RFM resistance and double resistance to RFM and TMP, were observed to increase significantly in simulated microgravity-grown B. subtilis but not in S. epidermidis. Objective 2. Launch to and recovery of BRIC-18 Flight (FL) samples from ISS. Performance of asynchronous Ground Control (GC) experiment and recovery of GC samples. Flight and Ground Controls. The BRIC-18 hardware payload was launched on the third SpaceX cargo resupply mission to the ISS (SpaceX-3) on 18 April 2014, using the Falcon 9 rocket and Dragon capsule configuration. The Dragon capsule docked with the ISS on 21 April 2014, then BRIC-A and BRIC-B canisters were transferred to the ISS on 22 April 2014 and stowed in the US laboratory module. Actuation of both canisters (i.e., injection of growth medium) was performed on 30 April 2014 at 10:08 EDT and samples were incubated at ISS ambient temperature until 5 May 2014 at 12:04 PM EDT, resulting in a total incubation time of 122 h. The experiment was terminated by transfer into the onboard -80°C MELFI freezer. The Dragon capsule returned to Earth on 18 May 2014 and samples were maintained in the frozen state until return to KSC for de-integration and further processing. Asynchronous GC experiments were performed in BRIC canisters in the ISS Experiment Simulation (ISSES) chamber at KSC according to the timeline and temperature profile recorded during the FL experiment. Following incubation, samples were transferred to a -80°C freezer and stored until further processing. Post-experiment sample processing. BRIC canisters were moved from storage at -80°C to a 4°C refrigerator and allowed to thaw overnight. Petri dishes were removed from PDFUs, cells resuspended using sterile disposable rubber spatulas, cultures transferred to sterile 50-mL conical centrifuge tubes, and the total volume recovered was measured. For viable counts, cultures were diluted serially tenfold in TSY medium, dilutions plated on TSY, and colonies counted after incubation at 37°C for 24 hours. To select for RFMR mutants, cultures were concentrated by centrifugation, plated without dilution onto TSY+RFM plates, and colonies counted after incubation at 37°C for 24 hours. The frequency of mutation to RFMR was calculated by dividing the total number of RFMR mutants by the total number of viable cells from each culture. Individual RFMR mutants were streak-purified and processed for further characterization by DNA sequencing. DNA sequencing and analyses. RFMR region of the B. subtilis and S. epidermidis rpoB genes were PCR amplified directly from cells and their nucleotide sequences determined at the University of Florida Interdisciplinary Center for Biotechnology (UF-ICBR). The rpoB sequences from FL and GC samples were aligned with the wild-type rpoB sequence using the online Clustal Omega server (http://www.ebi.ac.uk/) to identify the position of mutations relative to the sequences of the wild-type rpoB sequences obtained in parallel from B. subtilis strain WN1532 and S. epidermidis ATCC12228. This sequencing project is ongoing. We obtained a total of 330 RFMR mutants from both organisms and to date have sequenced 142 of them. The complete set of FL and GC results will be reported upon and published within the next year. Objective 3. Characterize altered phenotypes of the mutants obtained in Objectives (1) and (2). We have just recently begun to make progress on this objective during Summer of 2015. New graduate student Michael Morrison has calibrated the Omnilog Phenotype Microarray instrument to quantify the pattern of susceptibility of B. subtilis and S. epidermidis to a battery of 72 different antibiotics and growth-suppressing compounds, each delivered at 4 different concentrations. Results will be reported and published within the next year.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
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